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Roll No

EC-5002-CBGS

B.E. V Semester

Examination, December 2020

Choice Based Grading System (CBGS)

Digital Communication

Time : Three Hours

Maximum Marks : 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

1. a) Discuss about cumulative distribution function and probability density function.
b) Discuss Mean, Variance and Standard Deviation of random variable.
2. a) State and explain sampling Theorem of Low Pass Signal.
b) What are the various process involved in PCM? With the help of block diagram explain the working of PCM transmitter and receiver.
3. a) Write short notes on correlation between Random variable.
b) Discuss Flat Top Sampling. What is aperture effect and how it could be overcome?
4. a) Draw block diagram of PCM system and explain it in detail.
b) 24 telephone channels, each band limited to 3.4 kHz, are to be time division multiplexed by using PCM. Calculate the bandwidth of the PCM system for 128 quantization levels and an 8 kHz sampling frequency.

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5.
 - a) What is the major advantages of adaptive delta modulation over delta modulation? Draw block diagram of ADM system and explain it.
 - b) Write and explain differential PCM and vocoders.
6.
 - a) Explain about matched filter and correlator.
 - b) Discuss Bayes theorem.
7.
 - a) Consider a source with 8 messages having probabilities 0.22, 0.20, 0.18, 0.15, 0.10, 0.08, 0.05 and 0.02 respectively. Find the codes for messages using Huffman algorithm. Also calculate the efficiency of the codes.
 - b) Explain the information capacity theorem of channel coding.
8. Write short notes on any two
 - a) Central limits Theorem
 - b) Delta Modulation
 - c) Minimum Shift Keying (MSK)
 - d) Matched filter and Correlator
